

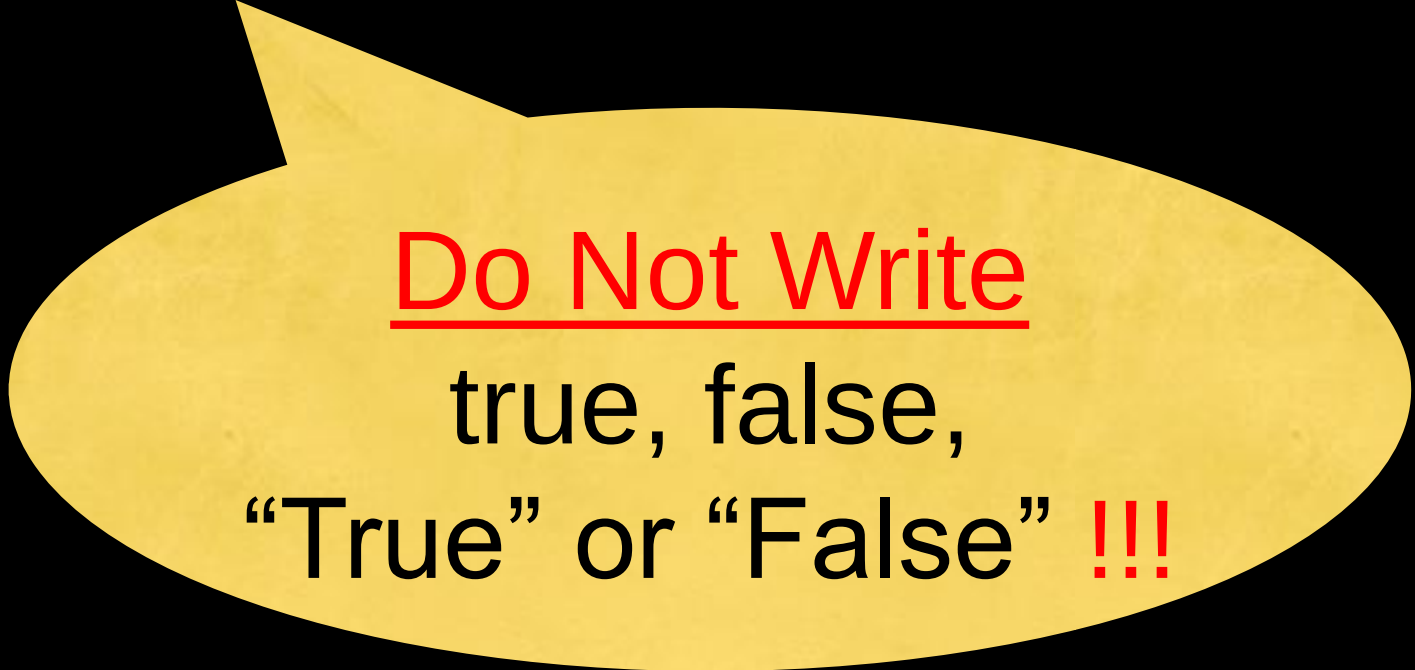
Lecture 1c

Basic Python

Boolean Truth Values

Boolean data type is named after the mathematician **George Boole**.

The **Boolean data type** has only two values :
`True` and `False`



Do Not Write
true, false,
"True" or "False" !!!

Comparison Operators

Operator	Meaning
<code>==</code>	Equal to
<code>!=</code>	Not equal to
<code><</code>	Less than
<code>></code>	Greater than
<code><=</code>	Less than or equal to
<code>>=</code>	Greater than or equal to

The Difference Between The == And = Operators

Operator	Meaning
==	(equal to) : asks whether two values are the same as each other.
=	(assignment) : puts the value on the right into the variables on the left. eg. <code>distance = 5</code>

Boolean Operators

Operator	Meaning
<code>and</code>	The <code>and</code> operator evaluates an expression to be <code>True</code> if both of the two Boolean values are <code>True</code> .
<code>or</code>	The <code>or</code> operator evaluates an expression to be <code>True</code> if either of the two Boolean values is <code>True</code> .
<code>not</code>	The <code>not</code> operator simply evaluates to the opposite of the Boolean value. eg. <code>not True</code> is <code>False</code>

The and Operator's Truth Table

Expression	Evaluates to ...
True and True	True
True and False	False
False and True	False
False and False	False

It must
be
False
once you
see a
False!

The `or` Operator's Truth Table

Expression	Evaluates to ...
True or True	True
True or False	True
False or True	True
False or False	False

It must
be True
once you
see a
True !

The `not` Operator's Truth Table

Expression	Evaluates to ...
<code>not True</code>	False
<code>not False</code>	<code>True</code> True

What is
`not not`
True ?

More about Boolean Truth Value

In Python 3.x **True** and **False** will always be equal to **1** and **0**

Anything that is **not 0** or **not empty** will also be evaluated as **True**